Intuitionistic Logics 6

6.1 Philosophical Motivations

Intuitionistic logic emerged in the early twentieth century as an alternative to classical logic. Its development was closely tied to foundational debates in mathematics, especially concerning the nature of mathematical truth, existence, and proof. Unlike classical logic, but like most three-valued logics, intuitionistic logic rejects the unrestricted validity of the law of excluded middle.

The philosophical roots of intuitionism lie primarily in the work of the Dutch mathematician and philosopher Luitzen Egbertus Jan Brouwer, whose programme sought to reconstruct mathematics on a purely constructive basis. Later, Arend Heyting (among others) formalised and extended these ideas into a system of logic usually regarded as intuitionistic logic.

Brouwer regarded mathematics not as a discovery of objective truths existing independently of human cognition, but as an activity of mental construction. According to him, mathematics originates in the basic intuition of time: the awareness of the passage from one moment to another. From this primordial intuition, the natural numbers and all mathematical constructions arise.

For Brouwer, mathematical objects do not exist independently of our capacity to construct them. Existence in mathematics therefore means constructibility. To claim that an object exists is to possess a method for explicitly constructing it. Moreover, to assert something, means that we have the means to prove it.

The most famous consequence of Brouwer's philosophy is the rejection of the law of excluded middle: $\ulcorner(\mathtt{A} \lor \neg\mathtt{A})\urcorner$. Classical logic treats this principle as universally valid. Brouwer argued, however, that such a principle is justified only when one possesses a method either to prove $\mathtt{A}$ or to refute it. In infinite domains, many propositions may remain undecided.

> Now the *principium tertii exclusi*: this demands that every supposition is either correct or incorrect, mathematically: that of every supposed fitting in a certain way of systems in one another, either the termination or the blockage by impossibility, can be constructed. The question of the validity of the *principium tertii exclusi* is thus equivalent to the question concerning the possibility of unsolvable mathematical problems. For the already proclaimed conviction that unsolvable mathematical problems do not exist, no indication of a demonstration is present.

> As long as only certain finite discrete systems are posited, the investigation into the possibility or impossibility of a fitting can always be terminated and leads to an answer whence the *principium tertii exclusi* is a reliable principle of reasoning. (Brouwer 2017, p. 42)

Brouwer therefore distinguished between:

- proving that a proposition is true,
- proving that it is false,
- and merely lacking either proof.

A proposition can be regarded as true only if there exists a construction proving it. As examples of statements about mathematics which Brouwer regarded as not certain to have a, he included (cf. Brouwer 2017, p. 44):

1. In the decimal expansion of π, there is 'a digit that occurs enduringly more often than all others'.
2. In the decimal expansion of π, 'there occur ... infinitely many pairs of equal consecutive digits'.

In sum, while in classical logic, truth is generally understood semantically: a statement is true if it corresponds to reality or satisfies a model; intuitionism instead identifies truth with provability aka demonstrability.

In this section we will present three systems related to intuitionistic logic: Hilbert's positive propositional logic P0, Ingebrigt Johansson's minimal propositional logic M0, and Heyting's intuitionistic propositional logic I0. The first of these is a system of logic without negation, on the basis of which many other logics can be built. The second adds negation, but

In this section we will present three systems related to intuitionistic logic: Hilbert's positive propositional logic P0, Ingebrigt Johansson's minimal propositional logic M0, and Heyting's intuitionistic propositional logic I0.

The first of these is a system of logic without negation, on the basis of which many other logic systems can be constructed. The second adds negation, but without validating neither the *ex contradictione quodlibet* nor the *tertium non datur*. Finally, Heyting's intuitionistic logic extends minimal logic by incorporating this principle, thereby yielding the standard propositional system associated with Brouwer's intuitionistic conception of mathematics.

6.2 Vocabularies and Languages

The vocabularies and formal languages $\mathbf{Fo}_{\mathsf{M0}}$ and $\mathbf{Fo}_{\mathsf{I0}}$ will be defined as those of S0.

Definition 6.1.

$$\mathbf{Vo}_{\mathsf{M0}} \stackrel{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}. \qquad \mathbf{Vo}_{\mathsf{I0}} \stackrel{\text{DF}}{=} \mathbf{Vo}_{\mathsf{S0}}.$$
$$\mathbf{Fo}_{\mathsf{M0}} \stackrel{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}. \qquad \mathbf{Fo}_{\mathsf{I0}} \stackrel{\text{DF}}{=} \mathbf{Fo}_{\mathsf{S0}}.$$

The vocabulary and language of P0, however, must be different, as there is no negation in this system.

Task 6.1. Define (a) '$\mathbf{Vo}_{\mathsf{P0}}$' and (b) '$\mathbf{Fo}_{\mathsf{P0}}$'.

Definition 6.2 (Vocabulary of P0 (answer to task 6.1.a))**.**

$$\mathbf{Log}_{\mathsf{P0}} \stackrel{\text{DF}}{=} \mathbf{Con} - \{`\neg'\},$$
$$\mathbf{Non}_{\mathsf{P0}} \stackrel{\text{DF}}{=} \mathbf{Non}_{\mathsf{S0}},$$
$$\mathbf{Aux}_{\mathsf{P0}} \stackrel{\text{DF}}{=} \mathbf{Par},$$
$$\mathbf{Vo}_{\mathsf{P0}} \stackrel{\text{DF}}{=} \left\langle \mathbf{Log}_{\mathsf{P0}}, \mathbf{Non}_{\mathsf{P0}}, \mathbf{Aux}_{\mathsf{P0}} \right\rangle .$$

Definition 6.3 (Formulae of P0 (solution to task 6.1.b))**.**

$$\mathbf{Fo}_{\mathsf{P0}} \stackrel{\text{DF}}{=} \mathbf{Non}_{\mathsf{P0}} \cup \{\ulcorner(\mathsf{p} \heartsuit \mathsf{q})\urcorner : \mathsf{p}, \mathsf{q} \in \mathbf{Fo}_{\mathsf{P0}} \mathbin{⩕} \heartsuit \in \mathbf{Log}_{\mathsf{P0}}\}.$$

6.3 Inference and Consequence Relations

P0, M0, and I0 fall among modern systems, so their inference relations are just special cases of definition 3.31.

As for the contents of their consequence relations, we want the law of excluded third to be not valid in general, as with the three-valued logics we have seen before. In other words, we do not want that all inferences of the following form be members of their consequence relations:

1. $/ \ (\mathtt{A} \lor \neg\mathtt{A})$

This will be necessarily the case for P0, since it has no negation symbol. For the other systems, we will need to construct their consequence relation so that this does law does not hold in general.

However, we do want, following Brouwer, that:

2. $/ \ \neg(\mathtt{A} \land \neg\mathtt{A})$
3. $(\mathtt{A} \rightarrow \mathtt{B}), (\mathtt{B} \rightarrow \mathtt{C}) \ / \ (\mathtt{A} \rightarrow \mathtt{C})$

Moreover, contrary to what happens in three-valued logics, we want that:

4. / $\neg\neg(\mathtt{A} \vee \neg\mathtt{A})$

This law makes sense in the context of intuitionistic logic, as it may be read as stating that there is no construction that turns a proof of $\ulcorner(\mathtt{A} \vee \neg\mathtt{A})\urcorner$ into a contradiction, or it is impossible to refute $\ulcorner(\mathtt{A} \vee \neg\mathtt{A})\urcorner$.

This is not necessarily against the principles of intuitionistic logic. Thus, intuitionism asserts that no proof of $\ulcorner(\mathtt{A} \vee \neg\mathtt{A})\urcorner$ exists for some cases: those cases in which no proof of neither $\mathtt{A}$ nor $\ulcorner\neg\mathtt{A}\urcorner$ is given. However, it does not assert that $\ulcorner\neg(\mathtt{A} \vee \neg\mathtt{A})\urcorner$. Moreover, it states that

Something else we want is all these to be *proper sublogics* of $\mathsf{S0}$. In other words, we want that:

$$\mathbf{Con}_{\mathsf{P0}}, \mathbf{Con}_{\mathsf{M0}}, \mathbf{Con}_{\mathsf{I0}} \subset \mathbf{Con}_{\mathsf{S0}}.$$

Moreover, we want that:

$$\mathbf{Con}_{\mathsf{P0}} \subset \mathbf{Con}_{\mathsf{M0}} \subset \mathbf{Con}_{\mathsf{I0}} \subset \mathbf{Con}_{\mathsf{S0}}.$$

We can show this if we show that all inferences valid in any of $\mathbf{Con}_{\mathsf{P0}}, \mathbf{Con}_{\mathsf{M0}}, \mathbf{Con}_{\mathsf{I0}}$ are also valid in $\mathbf{Con}_{\mathsf{S0}}$, and that, for each $\mathbf{Con}_{\mathsf{P0}}, \mathbf{Con}_{\mathsf{M0}}, \mathbf{Con}_{\mathsf{I0}}$, there is at least one inference that is $\mathbf{Con}_{\mathsf{S0}}$ but not in $\mathbf{Con}_{\mathsf{P0}}, \mathbf{Con}_{\mathsf{M0}}, \mathbf{Con}_{\mathsf{I0}}$, respectively.

6.4 Interpretation Systems

There are many interpretation systems for intuitionistic logics. Many of these require a possible worlds semantics, for which we would need at least to lessons to introduce. So we will focus on the derivation systems.

6.5 Derivation Systems

There are many derivation systems proposed for intuitionistic logics. It is a good idea to start by showing the axiomatic derivation systems which have been proposed for these systems, which are as follows.

Consider the following schematic inference forms:

$$\begin{aligned}
(\mathtt{A} \longrightarrow \mathtt{B}), \mathtt{A} \ &/ \ \mathtt{B} && \text{(MPP)}\\
&/ \ (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{A})) && \text{(P1)}\\
&/ \ ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C}))) && \text{(P2)}\\
&/ \ (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow (\mathtt{A} \land \mathtt{B}))) && \text{(P3)}\\
&/ \ ((\mathtt{A} \land \mathtt{B}) \longrightarrow \mathtt{A}) && \text{(P4)}\\
&/ \ ((\mathtt{A} \land \mathtt{B}) \longrightarrow \mathtt{B}) && \text{(P5)}\\
&/ \ (\mathtt{A} \longrightarrow (\mathtt{A} \lor \mathtt{B})) && \text{(P6)}\\
&/ \ (\mathtt{B} \longrightarrow (\mathtt{A} \lor \mathtt{B})) && \text{(P7)}\\
&/ \ ((\mathtt{A} \longrightarrow \mathtt{C}) \longrightarrow ((\mathtt{B} \longrightarrow \mathtt{C}) \longrightarrow ((\mathtt{A} \lor \mathtt{B}) \longrightarrow \mathtt{C}))) && \text{(P8)}\\
&/ \ ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A})) && \text{(M1)}\\
&/ \ (\neg\mathtt{A} \longrightarrow (\mathtt{A} \longrightarrow \mathtt{B})) && \text{(I1)}\\
&/ \ (\neg\neg\mathtt{A} \longrightarrow \mathtt{A}) && \text{(S1)}
\end{aligned}$$

In the context of these rules, we assume that $\ulcorner(\mathtt{A} \leftrightarrow \mathtt{B})\urcorner$ is an abbreviation of $\ulcorner((\mathtt{A} \longrightarrow \mathtt{B}) \land (\mathtt{B} \longrightarrow \mathtt{A}))\urcorner$.

From certain combinations these inference forms, it is possible to construct axiomatic deductive bases for P0, M0, I0, and even S0.

The axiomatic deductive base for P0 would consist of (MPP) plus (P1)–(P8). This is the basis for many deductive bases.

Thus, the axiomatic deductive base for M0 consists of P0's plus (M1). The axiomatic deductive base for I0 consists of M0's plus (I1). Finally, the axiomatic deductive base for S0 consists of I0's plus (S1).

This is sufficient definition of the deductive bases of these systems. However, we want to construct Fitch derivation systems, as they are more adequate representations of how reasoning works in the framework of these systems.

Derivation System for P0. The deductive base of the propositional positive logic P0 may be presented in the style of notation 3.48 as follows.

Reiteration.

$$\begin{array}{r|ll} a & \mathtt{A} & \\ & \vdots & \\ & \mid \ldots \mid \mathtt{A} & \text{Reit}, a \end{array}$$

Conjunction.

$$\begin{array}{r|ll} a & (\mathtt{A} \wedge \mathtt{B}) & \\ & \vdots & \\ & \mathtt{A} & \wedge^-_1, a \end{array} \qquad \begin{array}{r|ll} a & (\mathtt{A} \wedge \mathtt{B}) & \\ & \vdots & \\ & \mathtt{B} & \wedge^-_2, a \end{array} \qquad \begin{array}{r|ll} a & \mathtt{A} & \\ & \vdots & \\ b & \mathtt{B} & \\ & \vdots & \\ & (\mathtt{A} \wedge \mathtt{B}) & \wedge^+, a, b \end{array}$$

Disjunction.

$$\begin{array}{r|ll} a & (\mathtt{A} \vee \mathtt{B}) & \\ & \vdots & \\ b & \mid \underline{\mathtt{A}} & \\ & \mid \vdots & \\ c & \mid \mathtt{C} & \\ & \vdots & \\ d & \mid \underline{\mathtt{B}} & \\ & \mid \vdots & \\ e & \mid \mathtt{C} & \\ & \vdots & \\ & \mathtt{C} & \vee^-, a, b\text{–}c, d\text{–}e \end{array} \qquad \begin{array}{r|ll} a & \mathtt{A} & \\ & \vdots & \\ & (\mathtt{A} \vee \mathtt{B}) & \vee^+_1, a \end{array} \qquad \begin{array}{r|ll} a & \mathtt{B} & \\ & \vdots & \\ & (\mathtt{A} \vee \mathtt{B}) & \vee^+_2, a \end{array}$$

Conditional.

$$\begin{array}{r|ll} a & (\mathtt{A} \to \mathtt{B}) & \\ & \vdots & \\ b & \mathtt{A} & \\ & \vdots & \\ & \mathtt{B} & \to^-, a, b \end{array} \qquad \begin{array}{r|ll} a & \mid \underline{\mathtt{A}} & \\ & \mid \vdots & \\ b & \mid \mathtt{B} & \\ & \vdots & \\ & (\mathtt{A} \to \mathtt{B}) & \to^+, a\text{–}b \end{array}$$

Biconditional.

$$
\begin{array}{r|ll}
a & (\mathtt{A} \leftrightarrow \mathtt{B}) & \\
 & \vdots & \\
b & \mathtt{A} & \\
 & \vdots & \\
 & \mathtt{B} & \leftrightarrow^{-}_{1}, a, b
\end{array}
$$

$$
\begin{array}{r|ll}
a & (\mathtt{A} \leftrightarrow \mathtt{B}) & \\
 & \vdots & \\
b & \mathtt{B} & \\
 & \vdots & \\
 & \mathtt{A} & \leftrightarrow^{-}_{2}, a, b
\end{array}
$$

$$
\begin{array}{r|l|l}
a & & \underline{\mathtt{A}} \\
 & & \vdots \\
b & & \mathtt{B} \\
 & \vdots & \\
c & & \underline{\mathtt{B}} \\
 & & \vdots \\
d & & \mathtt{A} \\
 & \vdots & \\
 & (\mathtt{A} \leftrightarrow \mathtt{B}) \quad \leftrightarrow^{+}, a\text{–}b, c\text{–}d &
\end{array}
$$

Thus, we might define $\mathbf{DBas}_{\mathsf{P0}}$ as follows:

Definition 6.4 (Deductive base for P0).

$\mathbf{DBas}_{\mathsf{P0}} \overset{\text{DF}}{=} \{\mathsf{Reit}, \wedge^{+}, \wedge^{-}_{1}, \wedge^{-}_{2}, \vee^{+}_{1}, \vee^{+}_{2}, \vee^{-}, \rightarrow^{+}, \rightarrow^{-}, \leftrightarrow^{+}, \leftrightarrow^{-}_{1}, \leftrightarrow^{-}_{2}\}$.

As for the derivation system $\mathbf{Cal}_{\mathsf{P0}}$, we will assume that the functions $\mathbf{Prem}_{\mathsf{P0}}$ and $\mathbf{Conc}_{\mathsf{P0}}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.5 ($\mathbf{Cal}_{\mathsf{P0}}$).

$\mathbf{Cal}_{\mathsf{P0}}$ is a calculus on P0

$$
\begin{aligned}
&\overset{\text{DF}}{=} \mathbf{Cal}_{\mathsf{P0}} = \langle \mathbf{DBas}_{\mathsf{P0}}, \mathbf{Der}_{\mathsf{P0}}, \mathbf{Prem}_{\mathsf{P0}}, \mathbf{Conc}_{\mathsf{P0}}, \vdash_{\mathsf{P0}} \rangle \\
&\;: \mathbf{DBas}_{\mathsf{P0}} \text{ is a deductive base on } \mathbf{Fo}_{\mathsf{P0}} \ \bigwedge\!\!\!\!\bigwedge \\
&\quad \mathbf{Der}_{\mathsf{P0}} = \{\mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_{\mathsf{P0}} \text{ from } \mathbf{DBas}_{\mathsf{P0}}\} \ \bigwedge\!\!\!\!\bigwedge \\
&\quad \mathbf{Prem}_{\mathsf{P0}} : \mathbf{Der}_{\mathsf{P0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{P0}} \ \bigwedge\!\!\!\!\bigwedge \\
&\quad \mathbf{Conc}_{\mathsf{P0}} : \mathbf{Der}_{\mathsf{P0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{P0}} \ \bigwedge\!\!\!\!\bigwedge \\
&\quad \vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{P0}} \, (\mathbf{Prem}_{\mathsf{P0}}(\mathbf{X}) = \mathbf{A} \ \bigwedge\!\!\!\!\bigwedge \ \mathbf{Conc}_{\mathsf{P0}}(\mathbf{X}) = \mathbf{B})\}.
\end{aligned}
$$

Task 6.2. Show that all inferences of the forms (MPP), (P1)–(P8) (on p. 123) are members of $\vdash_{\mathsf{P0}}$.

Theorem 6.6. $\vdash_{\mathtt{P0}} ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C})))$.

Proof. Let $\mathtt{A}, \mathtt{B}, \mathtt{C} \in \mathbf{Fo}_{\mathtt{P0}}$ and consider the schematic derivation:

$$
\begin{array}{r|l|l}
1 & \quad \mid\ (\mathtt{A} \longrightarrow \mathtt{B}) & \\
2 & \quad \mid\ \quad \mid\ (\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) & \\
3 & \quad \mid\ \quad \mid\ \quad \mid\ \mathtt{A} & \\
4 & \quad \mid\ \quad \mid\ \quad \mid\ (\mathtt{B} \longrightarrow \mathtt{C}) & \longrightarrow^{-}, 2, 3 \\
5 & \quad \mid\ \quad \mid\ \quad \mid\ \mathtt{B} & \longrightarrow^{-}, 1, 3 \\
6 & \quad \mid\ \quad \mid\ \quad \mid\ \mathtt{C} & \longrightarrow^{-}, 4, 5 \\
7 & \quad \mid\ \quad \mid\ (\mathtt{A} \longrightarrow \mathtt{C}) & \longrightarrow^{+}, 3\text{–}6 \\
8 & \quad \mid\ ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C})) & \longrightarrow^{+}, 2\text{–}7 \\
9 & ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C}))) & \longrightarrow^{+}, 1\text{–}8
\end{array}
$$

Call this schematic derivation '$\mathbf{D}$'. It is clear, for all $\mathtt{A}, \mathtt{B}, \mathtt{C} \in \mathbf{Fo}_{\mathtt{P0}}$, that $\mathbf{D}$ is a Fitch derivation on $\mathbf{Fo}_{\mathtt{P0}}$ from $\mathbf{DBas}_{\mathtt{P0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathtt{P0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathtt{P0}}(\mathbf{D})$ and $\ulcorner((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C})))\urcorner \in \mathbf{Conc}_{\mathtt{P0}}(\mathbf{D})$. Therefore, $\vdash_{\mathtt{P0}} ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow (\mathtt{B} \longrightarrow \mathtt{C})) \longrightarrow (\mathtt{A} \longrightarrow \mathtt{C})))$ has to hold according to definition 6.5. □

Derivation System for M0. We want to present the deductive base for the minimal propositional logic M0 in the style of notation 3.48.

To do so, it is convenient (though not strictly necessary), to introduce the symbol '$\perp$', called 'the falsum'. This is not a symbol of M0's language (though it could have been introduced as such). However, we will have rules associated to this symbol which will allow us to obtain formulae of $\mathbf{Fo}_{\mathtt{M0}}$. Those rules are as follows:

Falsum.

$$
\begin{array}{r|ll}
a & \perp & \\
 & \vdots & \\
 & \mathtt{B} & \perp^{-}, a
\end{array}
\qquad\qquad
\begin{array}{r|ll}
a & \mathtt{A} & \\
 & \vdots & \\
b & \neg\mathtt{A} & \\
 & \vdots & \\
 & \perp & \perp^{+}, a, b
\end{array}
$$

We moreover rewrite the rules for negation as follows:

Negation.

$$\begin{array}{r|l|l l}
a & & \neg\mathtt{A} & \\
 & & \vdots & \\
b & & \bot & \\
 & \vdots & & \\
 & \mathtt{A} & & \neg^-, a\text{–}b
\end{array}
\qquad\qquad
\begin{array}{r|l|l l}
a & & \mathtt{A} & \\
 & & \vdots & \\
b & & \bot & \\
 & \vdots & & \\
 & \neg\mathtt{A} & & \neg^+, a\text{–}b
\end{array}$$

We will not use all these rules for neither M1 nor I1, though. We want to have negated sentences in both M1 and I1, so $\neg^+$ seems to be acceptable in both systems. In order to obtain a negated formula in these derivation systems, however, we also need a way to introduce the falsum, so $\bot^+$ will also be acceptable in both systems.

We nevertheless do not want to do *reductio ad absurdum* aka indirect proofs, so we will have to dispense with $\neg^-$ in both systems. In the particular case of minimal logic, we also do no want to have anything be derivable from contradictions, so we also want to dispense with $\bot^-$.

Thus, we might define $\mathbf{DBas}_{\mathsf{M0}}$ as follows:

Definition 6.7 (Deductive base for M0)**.**

$$\mathbf{DBas}_{\mathsf{M0}} \stackrel{\text{DF}}{=} \mathbf{DBas}_{\mathsf{P0}} \cup \{\neg^+, \bot^+\}.$$

As for the derivation system $\mathbf{Cal}_{\mathsf{M0}}$, we will once again assume that the functions $\mathbf{Prem}_{\mathsf{M0}}$ and $\mathbf{Conc}_{\mathsf{M0}}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.8 ($\mathbf{Cal}_{\mathsf{M0}}$)**.**

$\mathbf{Cal}_{\mathsf{M0}}$ is a calculus on M0

$$\begin{aligned}
&\stackrel{\text{DF}}{=} \mathbf{Cal}_{\mathsf{M0}} = \langle \mathbf{DBas}_{\mathsf{M0}}, \mathbf{Der}_{\mathsf{M0}}, \mathbf{Prem}_{\mathsf{M0}}, \mathbf{Conc}_{\mathsf{M0}}, \vdash_{\mathsf{M0}} \rangle \\
&\quad : \mathbf{DBas}_{\mathsf{M0}} \text{ is a deductive base on } \mathbf{Fo}_{\mathsf{M0}} \;\wedge\!\!\!\wedge \\
&\quad\;\; \mathbf{Der}_{\mathsf{M0}} = \{\mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_{\mathsf{M0}} \text{ from } \mathbf{DBas}_{\mathsf{M0}}\} \;\wedge\!\!\!\wedge \\
&\quad\;\; \mathbf{Prem}_{\mathsf{M0}} : \mathbf{Der}_{\mathsf{M0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{M0}} \;\wedge\!\!\!\wedge \\
&\quad\;\; \mathbf{Conc}_{\mathsf{M0}} : \mathbf{Der}_{\mathsf{M0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{M0}} \;\wedge\!\!\!\wedge \\
&\quad\;\; \vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{M0}} (\mathbf{Prem}_{\mathsf{M0}}(\mathbf{X}) = \mathbf{A} \;\wedge\!\!\!\wedge\; \mathbf{Conc}_{\mathsf{M0}}(\mathbf{X}) = \mathbf{B})\}.
\end{aligned}$$

Task 6.3. Show that all inferences of the forms (MPP), (P1)–(P8), (M1) (on p. 123) are members of $\vdash_{\mathsf{M0}}$.

It is clear that all inferences of the forms (P1)–(P8) are members of $\vdash_{\mathsf{M0}}$, since $\vdash_{\mathsf{M0}}$ is an extension of $\vdash_{\mathsf{P0}}$. Let us, then, prove the case of the inference form distinctive of M0, i.e., (M1).

Theorem 6.9. $\vdash_{\mathsf{M0}} ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A}))$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{M0}}$ and consider the schematic derivation:

$$
\begin{array}{r|l|l|l|ll}
1 & & (\mathtt{A} \longrightarrow \mathtt{B}) & & & \\ \cline{3-3}
2 & & & (\mathtt{A} \longrightarrow \neg\mathtt{B}) & & \\ \cline{4-4}
3 & & & & \mathtt{A} & \\ \cline{5-5}
4 & & & & \mathtt{B} & \longrightarrow^{-}, 1, 3 \\
5 & & & & \neg\mathtt{B} & \longrightarrow^{-}, 2, 3 \\
6 & & & & \bot & \bot^{+}, 4, 5 \\
7 & & & \neg\mathtt{A} & & \neg^{+}, 3\text{–}6 \\
8 & & ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A}) & & & \longrightarrow^{+}, 2\text{–}7 \\
9 & ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A})) & & & & \longrightarrow^{+}, 1\text{–}8
\end{array}
$$

Call this schematic derivation '$\mathbf{D}$'. It is clear, for all $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{M0}}$, that $\mathbf{D}$ is a Fitch derivation on $\mathbf{Fo}_{\mathsf{M0}}$ from $\mathbf{DBas}_{\mathsf{M0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathsf{M0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathsf{M0}}(\mathbf{D})$ and $\ulcorner((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A}))\urcorner \in \mathbf{Conc}_{\mathsf{M0}}(\mathbf{D})$. Therefore, by definition 6.8, $\vdash_{\mathsf{M0}} ((\mathtt{A} \longrightarrow \mathtt{B}) \longrightarrow ((\mathtt{A} \longrightarrow \neg\mathtt{B}) \longrightarrow \neg\mathtt{A}))$. □

Derivation System for I0. To present the deductive base $\mathbf{Cal}_{\mathsf{I0}}$ for the intuitionistic propositional logic I0 in the style of notation 3.48, we work from $\mathbf{Cal}_{\mathsf{M0}}$ and just add $\bot^{-}$. However, as we argued above, we should not add $\neg^{-}$ in order to avoid *reductio ad absurdum* proofs. Thus, we might define $\mathbf{DBas}_{\mathsf{I0}}$ as follows:

Definition 6.10 (Deductive base for I0)**.**

$$\mathbf{DBas}_{\mathsf{I0}} \overset{\text{DF}}{=} \mathbf{DBas}_{\mathsf{M0}} \cup \{\bot^{-}\}.$$

As for the derivation system $\mathbf{Cal}_{\mathsf{I0}}$, we will once again assume that the functions $\mathbf{Prem}_{\mathsf{I0}}$ and $\mathbf{Conc}_{\mathsf{I0}}$ map each Fitch derivation with their premises and conclusions, respectively. Thus, we have the following definition.

Definition 6.11 ($\mathbf{Cal}_{\mathsf{I0}}$)**.**

$\mathbf{Cal}_{\mathsf{I0}}$ is a calculus on $\mathsf{I0}$

$$
\begin{aligned}
&\overset{\mathrm{DF}}{=} \mathbf{Cal}_{\mathsf{I0}} = \langle \mathbf{DBas}_{\mathsf{I0}}, \mathbf{Der}_{\mathsf{I0}}, \mathbf{Prem}_{\mathsf{I0}}, \mathbf{Conc}_{\mathsf{I0}}, \vdash_{\mathsf{I0}} \rangle \\
&\quad : \mathbf{DBas}_{\mathsf{I0}} \text{ is a deductive base on } \mathbf{Fo}_{\mathsf{I0}} \wedge\!\!\wedge \\
&\quad\;\; \mathbf{Der}_{\mathsf{I0}} = \{\mathbf{D} : \mathbf{D} \text{ is a Fitch derivation on } \mathbf{Fo}_{\mathsf{I0}} \text{ from } \mathbf{DBas}_{\mathsf{I0}}\} \wedge\!\!\wedge \\
&\quad\;\; \mathbf{Prem}_{\mathsf{I0}} : \mathbf{Der}_{\mathsf{I0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{I0}} \wedge\!\!\wedge \\
&\quad\;\; \mathbf{Conc}_{\mathsf{I0}} : \mathbf{Der}_{\mathsf{I0}} \longrightarrow \wp\mathbf{Fo}_{\mathsf{I0}} \wedge\!\!\wedge \\
&\quad\;\; \vdash_{\mathsf{S}} = \{\langle \mathbf{A}, \mathbf{B} \rangle : \exists \mathbf{X} \in \mathbf{Der}_{\mathsf{I0}} (\mathbf{Prem}_{\mathsf{I0}}(\mathbf{X}) = \mathbf{A} \wedge\!\!\wedge \mathbf{Conc}_{\mathsf{I0}}(\mathbf{X}) = \mathbf{B})\}.
\end{aligned}
$$

Task 6.4. Show that all inferences of the forms (MPP), (P1)–(P8), (M1), (I1) (on p. 123) are members of $\vdash_{\mathsf{I0}}$.

It is clear that all inferences of the forms (P1)–(P8), (M1) are members of $\vdash_{\mathsf{I0}}$, since $\vdash_{\mathsf{I0}}$ is an extension of $\vdash_{\mathsf{M0}}$. Let us, then, prove the case of the inference form distinctive of $\mathsf{I0}$, i.e., (I1).

Theorem 6.12. $\vdash_{\mathsf{I0}} (\neg \mathtt{A} \longrightarrow (\mathtt{A} \longrightarrow \mathtt{B}))$.

Proof. Let $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{I0}}$ and consider the schematic derivation:

$$
\begin{array}{l|l|l}
1 & \quad \neg\mathtt{A} & \\
2 & \qquad \mathtt{A} & \\
3 & \qquad \bot & \bot^{+}, 2, 1 \\
4 & \qquad \mathtt{B} & \bot^{-}, 3 \\
5 & \quad (\mathtt{A} \longrightarrow \mathtt{B}) & \longrightarrow^{+}, 2\text{–}4 \\
6 & (\neg\mathtt{A} \longrightarrow (\mathtt{A} \longrightarrow \mathtt{B})) & \longrightarrow^{+}, 1\text{–}5
\end{array}
$$

Call this schematic derivation '$\mathbf{D}$'. It is clear, for all $\mathtt{A}, \mathtt{B} \in \mathbf{Fo}_{\mathsf{I0}}$, that $\mathbf{D}$ is a Fitch derivation on $\mathbf{Fo}_{\mathsf{I0}}$ from $\mathbf{DBas}_{\mathsf{I0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathsf{I0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathsf{I0}}(\mathbf{D})$ and $\ulcorner(\neg\mathtt{A} \longrightarrow (\mathtt{A} \longrightarrow \mathtt{B}))\urcorner \in \mathbf{Conc}_{\mathsf{I0}}(\mathbf{D})$. Therefore, by definition 6.11, $\vdash_{\mathsf{I0}} (\neg\mathtt{A} \longrightarrow (\mathtt{A} \longrightarrow \mathtt{B}))$. $\square$

An interesting fact about Heyting's intuitionistic logic is that, although we cannot have $\mathtt{A} \mathbin{/\!\!/} \neg\mathtt{A}$ in general, we do have $\neg\mathtt{A} \mathbin{/\!\!/} \neg\neg\neg\mathtt{A}$.

Task 6.5. Show that all inferences of the forms $\neg\mathtt{A} \mathbin{/\!\!/} \neg\neg\neg\mathtt{A}$ are members of $\vdash_{\mathsf{I0}}$.

In order to do task 6.5 we may prove each side independently.

Theorem 6.13. $\neg A \vdash_{I0} \neg\neg\neg A$

Proof. Let $A \in \mathbf{Fo}_{I0}$ and consider the schematic derivation:

$$
\begin{array}{r|lll}
1 & \neg A & & \\
\hline
2 & \quad | \; \neg\neg A & & \\
3 & \quad | \; \bot & & \bot^{+}, 1, 2 \\
4 & \neg\neg\neg A & & \neg^{+}, 2\text{–}3
\end{array}
$$

Call this schematic derivation '**D**'. It is clear, for all $A \in \mathbf{Fo}_{I0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{I0}$ from $\mathbf{DBas}_{I0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{I0}$ with $\{\ulcorner\neg A\urcorner\} \subseteq \mathbf{Prem}_{I0}(\mathbf{D})$ and $\ulcorner\neg\neg\neg A\urcorner \in \mathbf{Conc}_{I0}(\mathbf{D})$. Therefore, by definition 6.11, $\neg A \vdash_{I0} \neg\neg\neg A$. □

Theorem 6.14. $\neg\neg\neg A \vdash_{I0} \neg A$

Proof. Let $A \in \mathbf{Fo}_{I0}$ and consider the schematic derivation:

$$
\begin{array}{r|lll}
1 & \neg\neg\neg A & & \\
\hline
2 & \quad | \; A & & \\
3 & \quad | \quad | \; \neg A & & \\
4 & \quad | \quad | \; \bot & & \bot^{+}, 2, 3 \\
5 & \quad | \; \neg\neg A & & \neg^{+}, 3\text{–}4 \\
6 & \quad | \; \bot & & \bot^{+}, 5, 1 \\
7 & \neg A & & \neg^{+}, 2\text{–}6
\end{array}
$$

Call this schematic derivation '**D**'. It is clear, for all $A \in \mathbf{Fo}_{I0}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{I0}$ from $\mathbf{DBas}_{I0}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{I0}$ with $\{\ulcorner\neg\neg\neg A\urcorner\} \subseteq \mathbf{Prem}_{I0}(\mathbf{D})$ and $\ulcorner\neg A\urcorner \in \mathbf{Conc}_{I0}(\mathbf{D})$. Therefore, by definition 6.11, $\neg\neg\neg A \vdash_{I0} \neg A$. □

We can also prove, as desired, the double-negated variant of the *tertium non datur*.

Task 6.6. Show that all inferences of the form $/\!\!/\ \neg\neg(\mathtt{A} \vee \neg\mathtt{A})$ are members of $\vdash_{\mathrm{I0}}$.

Theorem 6.15. $\vdash_{\mathrm{I0}} \neg\neg(\mathtt{A} \vee \neg\mathtt{A})$

Proof. Let $\mathtt{A} \in \mathbf{Fo}_{\mathrm{I0}}$ and consider the schematic derivation:

1	$\quad\quad \neg(\mathtt{A} \vee \neg\mathtt{A})$	
2	$\quad\quad\quad \mathtt{A}$	
3	$\quad\quad\quad (\mathtt{A} \vee \neg\mathtt{A})$	$\vee^+_1, 2$
4	$\quad\quad\quad \bot$	$\bot^+, 3, 1$
5	$\quad\quad \neg\mathtt{A}$	$\neg^+, 2\text{–}4$
6	$\quad\quad (\mathtt{A} \vee \neg\mathtt{A})$	$\vee^+_2, 5$
7	$\quad\quad \bot$	$\bot^+, 1, 6$
8	$\quad \neg\neg(\mathtt{A} \vee \neg\mathtt{A})$	$\neg^+, 1\text{–}7$

Call this schematic derivation '**D**'. It is clear, for all $\mathtt{A} \in \mathbf{Fo}_{\mathrm{I0}}$, that **D** is a Fitch derivation on $\mathbf{Fo}_{\mathrm{I0}}$ from $\mathbf{DBas}_{\mathrm{I0}}$. This means that there is a derivation $\mathbf{D} \in \mathbf{Der}_{\mathrm{I0}}$ with $\{\} \subseteq \mathbf{Prem}_{\mathrm{I0}}(\mathbf{D})$ and $\ulcorner\neg(\mathtt{A} \vee \neg\mathtt{A})\urcorner \in \mathbf{Conc}_{\mathrm{I0}}(\mathbf{D})$. Therefore, by definition 6.11, $\vdash_{\mathrm{I0}} \neg(\mathtt{A} \vee \neg\mathtt{A})$. □